

D492

Data Analytics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A retailer's analytics team is designing a warehouse for fast monthly sales reports across 250 stores. The source order system uses a fully normalized design that took two years to build. Which design fits the reporting warehouse best?

- A. One flat table with every column, which is always fastest.
- B. A copy of the normalized order-system design, since it took so long to build.
- C. Normalize further to reduce every bit of redundancy.
- D. A star schema: a central sales fact table joined to denormalized dimensions such as store, product, and date.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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2.

A marketing analyst has a customers table with 5,000 rows and an orders table. 1,200 customers have never ordered. She wants a list of every customer, with order totals where they exist. The orders table was migrated from an older system in 2021. Which join should she use?

- A. customers CROSS JOIN orders
- B. customers LEFT JOIN orders
- C. customers INNER JOIN orders
- D. orders LEFT JOIN customers

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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3. [TRANSFER — reversed stem / different variable]

An INNER JOIN between a customers table (5,000 rows) and a loyalty sign-ups table returns 3,100 rows, one per matched customer. How many customers have no loyalty sign-up?

- A. 1,900
- B. 3,100
- C. 8,100
- D. It cannot be determined without running a LEFT JOIN.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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4.

An A/B test of a new checkout page at an online bookstore with 90,000 monthly visitors returns $p = 0.03$. The team set $\alpha = 0.05$ before the test. The designer spent three weeks on the new layout. What is the correct decision?

- A. Accept the new page as proven better with 97% certainty.
- B. Fail to reject the null hypothesis.
- C. Reject the null hypothesis; the difference is statistically significant.
- D. Rerun the test because three weeks of design work was involved.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A test run with $\alpha = 0.01$ returns $p = 0.04$. What is the decision?

- A. Accept the null as proven true.
- B. Fail to reject the null hypothesis.
- C. Reject the null, because p is above α .
- D. Reject the null, because p is below 0.05.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A fraud model at a credit card company with 3 million cardholders flags a legitimate purchase as fraud and blocks it. The null hypothesis is that each purchase is legitimate. The company's call center is in Tampa. What kind of error occurred?

- A. No error, because blocking is always the safer choice.
- B. Type II error (false negative).
- C. Sampling error.
- D. Type I error (false positive).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A screening test misses a patient who does have the condition. The null hypothesis is “no condition.” Which error is this?

- A. A false positive.
- B. Type I error.
- C. Type II error.
- D. Selection bias.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A hospital's analytics team, which just moved into a new office, must show the board how average ER wait time changed month by month over the last three years. Which chart fits best?

- A. A scatter plot of wait time against patient age.
- B. A pie chart.
- C. A line chart.
- D. A 3-D bar chart, since it is always the most engaging.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A utility's customer file of 250,000 records stores the same customer three times under slightly different spellings. The file sits on a server that was upgraded last month. Which data quality dimension is failing?

- A. Uniqueness.
- B. Validity.
- C. Timeliness, because the server changed recently.
- D. Completeness.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Delivery times for a courier have $Q1 = 20$ minutes and $Q3 = 40$ minutes. The median is 28, and the company has 35 drivers. Using the $1.5 \times IQR$ rule, above what time is a delivery an outlier?

- A. 70 minutes
- B. 50 minutes
- C. 60 minutes
- D. 58 minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Using the $1.5 \times \text{IQR}$ rule, a data set's lower outlier fence is 10 and Q1 is 25. What is Q3?

- A.** 40
- B.** 35
- C.** 15
- D.** 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

At an insurance company, a business-side analyst is assigned to maintain the definition of “active policy,” resolve policy-data quality issues, and enforce naming standards. She has worked there for nine years. What governance role is this?

- A.** Chief data officer, since only executives may define terms.
- B.** Data custodian.
- C.** Data owner.
- D.** Data steward.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A retailer's analytics team is designing a warehouse for fast monthly sales reports ~~across 250 stores~~. The source order system uses a fully normalized design ~~that took two years to build~~. Which design fits the reporting warehouse best?

Target word: warehouse design **Decides:** The purpose is fast analytic reporting.

A	Absolute language	Wide flat tables repeat data and are not always faster.
B	Camouflage	Build effort does not make a 3NF design suited to reporting.
C	Inverted operation	More normalization means more joins and slower reports.
D	Correct	Dimensional models are built for fast, simple reporting queries.

2. Answer B Concept 2

A marketing analyst has a customers table with 5,000 rows and an orders table. 1,200 customers have never ordered. She wants a list of every customer, with order totals where they exist. ~~The orders table was migrated from an older system in 2021~~. Which join should she use?

Target word: join type **Decides:** Every customer must appear, including non-buyers.

A	Confusion set	Pairs every customer with every order.
B	Correct	Keeps all 5,000 customers; non-buyers get NULL totals.
C	Dropped element	Drops the 1,200 customers with no orders.
D	Inverted operation	Keeps every order, not every customer.

3. Answer A Concept 2 · transfer

An INNER JOIN between a customers table (5,000 rows) and a loyalty sign-ups table returns 3,100 rows, one per matched customer. How many customers have no loyalty sign-up?

Target word: unmatched count from an inner join — reverse **Decides:** Inner join keeps only matches.

A	Correct	$5,000 - 3,100 = 1,900$ unmatched customers.
B	Wrong stem number	That is the matched count.
C	Inverted operation	Adds instead of subtracting.
D	Absolute language	With one row per match, subtraction gives the answer.

4. Answer C Concept 3

An A/B test of a new checkout page at an online bookstore ~~with 90,000 monthly visitors~~ returns $p = 0.03$. The team set $\alpha = 0.05$ before the test. ~~The designer spent three weeks on the new layout~~. What is the correct decision?

Target word: decision **Decides:** $p = 0.03 \leq \alpha = 0.05$.

A	Absolute language	A p-value is not the probability the alternative is true.
B	Inverted operation	Reverses the decision rule.
C	Correct	p is below α .
D	Camouflage	Design effort does not affect the statistical decision.

5. Answer B Concept 3 · transfer

A test run with $\alpha = 0.01$ returns $p = 0.04$. What is the decision?

Target word: decision at a stricter alpha **Decides:** $0.04 > 0.01$.

A	Absolute language	Failing to reject is not proof of the null.
B	Correct	p is above the stated alpha.
C	Inverted operation	Flips the rule.
D	Wrong stem number	Uses the habitual 0.05 instead of the stated 0.01.

6. Answer D Concept 4

A fraud model at a credit card company ~~with 3 million cardholders~~ flags a legitimate purchase as fraud and blocks it. The null hypothesis is that each purchase is legitimate. ~~The company's call center is in Tampa.~~ What kind of error occurred?

Target word: error type **Decides:** A true null (legitimate) was rejected.

A	Absolute language	A wrong block is still an error.
B	Inverted operation	Type II is missing real fraud.
C	Adjacent category	Sampling error is about estimates varying by sample, not this decision.
D	Correct	Rejecting a true null is a Type I error.

7. Answer C Concept 4 · transfer

A screening test misses a patient who does have the condition. The null hypothesis is “no condition.” Which error is this?

Target word: error type for a miss **Decides:** A false null was not rejected.

A	Confusion set	A miss is a false negative.
B	Inverted operation	Type I is a false alarm.
C	Correct	Failing to reject a false null is Type II.
D	Adjacent category	A sampling issue, not a decision error.

8. Answer C Concept 5

A hospital's analytics team, ~~which just moved into a new office,~~ must show the board how average ER wait time changed month by month over the last three years. Which chart fits best?

Target word: chart type **Decides:** Change over time.

A	Adjacent category	Shows a relationship, not a monthly trend.
B	Confusion set	Pies show parts of one whole, not trends.
C	Correct	Lines show trends across ordered time.
D	Absolute language	3-D distorts values and engagement is not the goal.

9. Answer A Concept 6

A utility's customer file of 250,000 records stores the same customer three times under slightly different spellings. ~~The file sits on a server that was upgraded last month.~~ Which data quality dimension is failing?

Target word: quality dimension **Decides:** The same entity appears more than once.

A	Correct	Duplicate records break uniqueness.
B	Adjacent category	Validity checks whether values follow format rules.
C	Camouflage	The upgrade does not create duplicate records.
D	Confusion set	Completeness is about missing values.

10. Answer A Concept 7

Delivery times for a courier have $Q1 = 20$ minutes and $Q3 = 40$ minutes. ~~The median is 28, and the company has 35 drivers.~~ Using the $1.5 \times IQR$ rule, above what time is a delivery an outlier?

Target word: upper fence **Decides:** $IQR = 40 - 20 = 20$; fence = $Q3 + 1.5 \times 20$.

A	Correct	$40 + 30 = 70$.
B	Confusion set	Adds 30 to $Q1$, the base for the lower fence.
C	Dropped element	$40 + 20$ drops the 1.5 multiplier.
D	Wrong stem number	Adds 30 to the median instead of $Q3$.

11. Answer B Concept 7 · transfer

Using the $1.5 \times IQR$ rule, a data set's lower outlier fence is 10 and $Q1$ is 25. What is $Q3$?

Target word: solve for $Q3$ **Decides:** $25 - 1.5 \times IQR = 10$, so $IQR = 10$.

A	Dropped element	Uses 15 as the IQR , dropping the division by 1.5.
B	Correct	$IQR = 15 / 1.5 = 10$; $Q3 = 25 + 10 = 35$.
C	Inverted operation	Subtracts the IQR from $Q1$ instead of adding it.
D	Wrong stem number	The lower fence, not $Q3$.

12. Answer D Concept 8

At an insurance company, a business-side analyst is assigned to maintain the definition of "active policy," resolve policy-data quality issues, and enforce naming standards. ~~She has worked there for nine years.~~ What governance role is this?

Target word: governance role **Decides:** Day-to-day care of definitions and quality.

A	Absolute language	Stewards define terms under governance oversight.
B	Confusion set	Custodians run storage and technical controls.
C	Adjacent category	The owner is the accountable executive for the domain.
D	Correct	Stewards maintain definitions, quality, and standards.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Normalized vs. dimensional design	Operational systems normalize (3NF) to avoid update errors; analytic warehouses use star schemas (a fact table plus denormalized dimensions) for fast reporting.	Copying the transactional 3NF design into the reporting warehouse.	“Fast reports” or “warehouse” vs. “order entry.”
2	INNER vs. LEFT JOIN	An INNER JOIN keeps only matching rows; a LEFT JOIN keeps every row from the left table and fills non-matches with NULL.	Using INNER JOIN when unmatched records must stay in the result.	“Every customer, including those with no orders.”
3	p-value decision rule	Reject the null hypothesis when $p \leq \alpha$; otherwise fail to reject it (never “accept” it).	Comparing p to 0.05 by habit when a different alpha was set.	A p-value and a stated alpha.
4	Type I vs. Type II error	Type I = rejecting a true null (false positive); Type II = failing to reject a false null (false negative).	Swapping the two.	A stated null plus a wrong flag or a miss.
5	Matching chart to purpose	Line charts show change over time; bar charts compare categories; scatter plots show relationships; pie charts only show parts of one whole.	Choosing the most eye-catching chart.	“Over time,” “compare,” or “relationship.”
6	Data quality dimensions	Completeness (no gaps), accuracy (matches reality), consistency (agrees across systems), timeliness (current), validity (fits rules), uniqueness (no duplicates).	Calling every quality problem “accuracy” or “completeness.”	The specific symptom: duplicates, blanks, stale values, bad formats.
7	IQR outlier rule	$IQR = Q3 - Q1$; outliers fall below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.	Adding $1.5 \times IQR$ to the median, or forgetting the 1.5.	Q1 and Q3 with a question about outlier limits.
8	Data governance roles	The data owner is accountable for a data domain; the data steward maintains definitions, quality, and standards day to day; the custodian runs the technical storage.	Calling the day-to-day definition keeper the owner or the custodian.	Maintaining business definitions and resolving quality issues.

Course scope (public description): Applications - Data Analytics - Applications covers advanced concepts across the various phases of the data product lifecycle. You will learn to choose and apply appropriate techniques for data management and data manipulation, statistical analysis, visualization, and data governance concepts to satisfy business needs.